12+ YEARS INDUSTRY EXPERIENCE • PRODUCTION PLANNING & SCM

Automate Your Excel Work. Turn Your Data Into Action.

Python Automation • Excel Automation • Streamlit Dashboards • Business Reporting

I don't just write scripts — I understand the business process behind the spreadsheet. 12+ years of manufacturing operations combined with modern Python pipelines to build bulletproof business automation.

1-Click Workflows | Zero Human Copy-Paste | Deterministic Validation | Shopfloor Streamlit Web Apps

Explore Case Studies → | Start a Project

Applied expertise from operations at: Tata Motors • Rheinmetall Automotive

12+ YEARS MANUFACTURING EXPERIENCE / PRODUCTION PLANNING & CONTROL (PPC) / SUPPLY CHAIN MANAGEMENT (SCM) / SAP PP / MM CERTIFIED / PYTHON AUTOMATION

CAPABILITIES

What I Automate

Replacing manual spreadsheet friction with robust Python automation engines and interactive decision-making dashboards.

6 Core Automation Modules

01

Python Excel Automation

Convert repetitive multi-file Excel processes into clean, one-click automated Python scripts.

CLIENT OUTCOME
Eliminates 90%+ of manual copy-paste time while guaranteeing 100% mathematical consistency.

- Multi-file workbook consolidation
- Automated VLOOKUP & cross-sheet joins
- Data cleaning & outlier removal
- Pre-formatted management exports

02

Automated Reporting

Turn tedious daily, weekly, and monthly reporting routines into instant automated report generators.

CLIENT OUTCOME
Reports that previously took hours across shifts now generate in under 5 seconds.

- Daily production & shift summaries
- Weekly executive status decks
- Monthly management KPI decks
- Automated PDF & Excel distribution

03

Streamlit Dashboards

Transform static spreadsheets into interactive, browser-based web applications with live filters and analytics.

CLIENT OUTCOME
Gives shopfloor planners and leaders real-time visibility into production status.

- Live shortage & float tracking
- Interactive line capacity planners
- Real-time Plan vs Actual visualizer
- Multi-stage pipeline aging analytics

04

Data Processing & Analysis

05

Business Workflow Automation

06

Manufacturing / SCM / PPC Automation

Clean, validate, structure, and reconcile large, messy datasets from disconnected systems.

CLIENT OUTCOME

Reliable, audit-ready data pipelines that handle growing volume without extra human effort.

BOM completeness auditing

Multi-system data reconciliation

Data anomaly & error detection

Automated schema transformation

Automation

Map, analyze, and automate multi-step operational workflows to eliminate organizational bottlenecks.

CLIENT OUTCOME

Streamlined business operations with predictable execution and zero manual bottlenecks.

End-to-end workflow automation

Multi-department data handoffs

Scheduled pipeline execution

Exception alerting & notifications

Automation

Specialized scheduling, capacity planning, and supply chain tracking tailored specifically for automotive & manufacturing.

CLIENT OUTCOME

Optimized assembly line balancing, reduced inventory holding, and minimized line starvation.

Finite capacity multi-line scheduling

Setup & changeover loss calculation

FIFO clear-to-build allocation

Paint shop buffer & aging tracking

BUSINESS TRANSFORMATION

The Shift: Manual vs. Automated

Comparing the high-risk friction of manual spreadsheet work against the reliability of automated Python systems.

Target: Hours of friction → Seconds of execution

MANUAL SPREADSHEET PROCESS (BEFORE)

High Error Risk

Multiple Disparate Workbooks 01
Raw dumps from ERP, MES, and local spreadsheets in mismatched formats

Manual Copy-Pasting 02
Planners spend hours copying rows, stitching tabs, and formatting layouts

Fragile VLOOKUPs & Formulas 03
Formula breakages (#N/A, #REF!) when columns shift or row counts change

Unnoticed Calculation Errors 04
Fatigue causes missed shortages, unspotted duplicate VINs, and inventory gaps

Delayed Decision Making 05
Shift reports take hours to produce, delaying critical line decisions

AUTOMATED PYTHON PIPELINE (AFTER)

100% Deterministic

1-Click Multi-File Upload 01
Instant parsing of .xlsx, .xlsb, and CSV dumps without manual touch

Automated Python Pipeline 02
Vectorized in-memory Pandas cleaning and joins executed in milliseconds

Automated BOM & Schema Audit 03
Deterministic validation catches missing parts and anomalies automatically

Automated Business Rules 04
FIFO clear-to-build sequencing, setup loss deductions, and holiday logic

Live Dashboard & Formatted Excel 05
Executive status decks, formatted Excel plans, and real-time Streamlit views

ENGINEERING CASE STUDIES

Interactive Production Case Studies

Production scheduling algorithms, real-time shopfloor shortage monitors, and Python Excel pipelines built to eliminate manual friction in manufacturing.

All 7 | Streamlit Dashboards 4 | Excel Automation 2 | PPC & SCM Engines 1 vin-generation-dashboa… Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | STREAMLIT DASHBOARD

VIN Generation PPC Dashboard

Enterprise-grade production planning dashboard for automotive assembly lines, managing vehicle scheduli...

Impact: 3.5 hrs → 8 sec
Shift allocation runtime reduced with 100% BO...

OPERATIONAL PROBLEM:
Scheduling vehicle drops onto assembly lines without real-time component stock verification...

TECHNICAL SOLUTION:
Built a live Streamlit dashboard with dynamic shift inventory tracking, FIFO clear-to-build allocation...

Python · Streamlit · Pandas · NumPy · OpenPyXL

App Preview | Case Study ↗

shortage-float-tracker… Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | STREAMLIT DASHBOARD

PPC Shortage & Float Tracker

Real-time component shortage monitoring system tracking wiring harnesses, cockpit assemblies, engines...

Impact: 11 Float Stages
Automated 6:30 AM shift reset across 4 critical...

OPERATIONAL PROBLEM:
Manual spreadsheet monitoring of component shortages across shifts is error-prone, slow, and fai...

TECHNICAL SOLUTION:
Built a 6-tab Streamlit dashboard with automated PBS/Sealant/Total Float shortage calculations, BO...

Python · Streamlit · Pandas · NumPy · OpenPyXL

App Preview | Case Study ↗

vin-production-plan.app Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | PPC & SCM ENGINE

VIN Generation & Production Plan

Multi-day rolling production sequencing system with automated float matching, variant constraint...

Impact: 3-4 Day Horizon
Rolling multi-stage vehicle sequencing across ...

OPERATIONAL PROBLEM:
Manual scheduling across BIW, Paint Shop, and TCF assembly stages leads to bottlenecks, float...

TECHNICAL SOLUTION:
Automated 3-4 day rolling production planning with multi-stage float prioritization (Paint Floor → WIP →...

Python · Streamlit · OpenPyXL

App Preview | Case Study ↗

datewise-planning.app Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | EXCEL AUTOMATION

Datewise Production Planning System

Finite capacity scheduling system for automotive component manufacturing, allocating daily production...

Impact: 13 Lines Scheduled
Setup loss deducted automatically (240m majo...

OPERATIONAL PROBLEM:
Manual day-wise production scheduling across 13 manufacturing lines doesn't account for setup...

TECHNICAL SOLUTION:
Built a capacity-aware scheduling engine that calculates setup time deductions, performs...

Python · Streamlit · Pandas · NumPy · XlsxWriter

App Preview | Case Study ↗

paint-aging-analytics… Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | STREAMLIT DASHBOARD

Paint Shop Aging Analytics

Multi-stage WIP aging analysis system for paint shop operations, tracking vehicle bodies from BIW through...

Impact: 100% WIP Visibility
Multi-stage BIW → PT → PBS aging analytics ...

OPERATIONAL PROBLEM:
Vehicle bodies in paint shop stages experience aging due to production delays or quality holds, but...

TECHNICAL SOLUTION:
Built an interactive aging report that classifies vehicles into standardized aging buckets (1 Day to...

Python · Streamlit · Pandas

App Preview | Case Study ↗

float-reports-portal.a… Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | STREAMLIT DASHBOARD

PPC Float Reports Portal

Automated vehicle float tracking and taper analysis portal, reconciling daily float data across assembly line...

Impact: 12 Files Reconciled
Dynamic multi-day float taper discovery gener...

OPERATIONAL PROBLEM:
Daily collation and trend analysis of vehicle float reports across TCF1 and TCF2 lines requires...

TECHNICAL SOLUTION:
Built a smart file-mapping dashboard that automatically identifies file roles and dates from...

Python · Streamlit · Pandas

App Preview | Case Study ↗

shiftwise-planner.app Internal Engine ↗

LATENCY < 450ms | VALIDATION 100% Pass | STATUS Active

Representative Study | EXCEL AUTOMATION

ShiftWise Production Planner

Daily production planning tool that allocates targets across 3 operating shifts with cycle time calculations a...

Impact: 3-Shift Waterfall
PPM-based cycle time calculation with zero m...

OPERATIONAL PROBLEM:
Allocating daily production targets across 3 shifts while accounting for line cycle times, changeovers,...

TECHNICAL SOLUTION:
Built a waterfall shift allocation engine that sequentially fills shift capacity based on parts-per-...

Python · Streamlit · Pandas · XlsxWriter

App Preview | Case Study ↗

ARCHITECTURE

Excel Automation Engine Blueprint

How custom Python automation transforms fragile, manual spreadsheets into reliable, 1-click execution pipelines.

Deterministic Workflow Engine

STAGE 01 · 01

Disparate Input Ingestion

dumps (.xlsx), MES float extracts (.xlsb), and supplier stock files without manual file opening.

- Multi-tab extraction
- Header schema alignment
- pyxlsb byte decompression
- Date index standardization

Flows to Stage 02 →

STAGE 02 · 02

Vectorized Processing

execute joins and aggregations in-memory across 50,000+ rows in milliseconds.

- BOM reconciliation
- Cross-worksheet matching
- Zero formula lag (#N/A free)
- Outlier cleansing

Flows to Stage 03 →

major/minor changeover deductions, 3-shift splits, and holiday calendars.

- 13-Line capacity balance
- Setup loss deduction (240m/60m)
- FIFO line allocation
- Shift auto-reset logic

Flows to Stage 04 →

sheets with corporate styling, column auto-fit, and visual KPI status.

- Executive summary tab
- Linewise production schedule
- Automated PDF generation
- Pre-formatted print views

Deliverable Ready

1-Click Execution & Zero Maintenance
Ingests raw transactional workbooks ➔ Runs validation rules ➔ Outputs executive deliverables in seconds.

Automate Your Workflow

CAREER & DOMAIN TRACK RECORD

12+ Years Industry Experience

Proven background across automotive assembly, supplier logistics, production scheduling, SAP ERP systems, and Python process automation.

• CURRENT PROFESSIONAL ENGAGEMENT (~3 MONTHS)

SCM / PPC / ORDER FULFILLMENT

Tata Motors Passenger Vehicle Ltd

Pune, India · Current (Ongoing)

Managing end-to-end supply chain planning, production aggregate sequencing, and order fulfillment workflows for passenger vehicle assembly lines.

KEY OPERATIONS & RESPONSIBILITIES

- Aggregate vehicle sequencing across Trim, Chassis, and Final (TCF) assembly tracks
- Real-time shortage identification and multi-channel component stock tracking
- Coordination with cross-functional shopfloor teams (BIW, Paint Shop, TCF) to prevent line starvation
- Development of custom automated Python utilities to accelerate daily PPC reporting

Supply Chain Management | PPC & SCM | Order Fulfillment | Line Sequencing | Python Utilities

Rheinmetall Automotive India PVT LTD

May 2018 – Apr 2026 (~8 years) · Pune, India

BU Bearings (formerly KSPG Automotive)

Assistant Manager – Production Planning & Control

- Developed and managed detailed Master Production Schedules (MPS) aligning sales forecasts with plant capacity
- Analyzed raw material requirements and performed regular MRP runs in SAP PP/MM with supplier coordination
- Oversaw material movement tracking in SAP (transfers, issues, receipts) ensuring 99%+ inventory accuracy
- Formulated inventory strategies that significantly reduced excess stock and increased turnover ratio
- Spearheaded PPC automation by developing AI-driven scheduling models for daily, weekly, and monthly planning
- Built dynamic Plan vs. Actual dashboard to monitor real-time production variances and line bottlenecks
- Awarded for Excellence in Digital Transformation Using AI (Feb 2026)

Production Planning | MPS & MRP | SAP PP/MM | Raw Material Planning | AI-Driven Planning | Plan vs Actual Analytics

- Managed daily aggregate sequencing for Xenon Trim, Chassis, and Final (TCF) lines
- Monitored TCF line constraints and provided real-time feedback for seamless trim-to-rollout flow
- Partnered with Production, Quality, and Logistics to eliminate line bottlenecks and ensure on-time dispatch
- Operated SAP IPMS and aggregate scanning systems across shifts
- Followed up with suppliers and conducted on-site vendor visits to resolve critical part shortages

Line Sequencing | SAP IPMS | Cross-Functional Coordination | Vendor Management | WCQ / SLT Standards

Education

BBA – Specialization in IT

Diploma in Mechanical

RMCET, Ratnagiri

Jun 2010 – May 2013

Certifications

SAP PP / MM

Pune Institute of SAP

May 2019 – Oct 2019

Award

Excellence in Digital Transformation Using AI

February 2026

PPC/SCM reporting to fully automated, AI-driven workflows and real-time analytics.

01

Understand

Understand the existing manual workflow, data sources, and operational pain points.

Stage 1 of 4

02

Analyze

Identify repetitive work, errors, data bottlenecks, and automation opportunities.

Stage 2 of 4

03

Automate

Build Python, Excel, or Streamlit automation with robust validation rules.

Stage 3 of 4

04

Deliver

Provide a simple, reliable solution with clear documentation and support.

Project Handover

Schedule a Workflow Review →

CORE COMPETENCIES

Skills & Technical Stack

Deep manufacturing shopfloor domain knowledge combined with modern Python automation, SAP ERP workflows, and Streamlit data applications.

Business Domain | ERP | Automation | Data & Dashboards | AI & Reporting

BUSINESS DOMAIN STACK & MODULES

- Production Planning & Control (PPC)
- Supply Chain Management (SCM)
- Master Production Schedule (MPS)
- Material Requirement Planning (MRP)
- Inventory Management
- Order Fulfillment
- Capacity Planning

BACKGROUND

Shopfloor Engineering Meets Python Automation

A unique dual-perspective background combining 12+ years of manufacturing operations with custom digital automation.

Most developers only see the data schema. Having spent over 12 years inside automotive assembly plants, bearing manufacturing lines, and PPC control rooms, I understand the real-world operational context behind the spreadsheet — line starvation risks, supplier stock variances, changeover losses, and shift handovers.

Whether managing daily aggregate vehicle sequencing at **Tata Motors Passenger Vehicle Ltd** or orchestrating Master Production Schedules (MPS) and SAP MRP runs at **Rheinmetall Automotive**, I build automation tools that solve actual shopfloor bottlenecks.

```
    plan = balance_13_lines(
        df_floats,
        setup_deductions={"major": 240, "minor": 60}
    )
```

12+	100%
YEARS OPERATIONS EXPERIENCE	DETERMINISTIC PIPELINE CODE

```
return plan.to_excel_styled("Daily_PPC_Plan.xlsx")
```

Pandas • NumPy • OpenPyXL   Execution: 420ms

Have a Manual Process That Should Be Automated?

Calculate your potential manual hours and cost savings below, then submit your workflow for a free technical feasibility review.

Automation ROI Calculator

Estimate time & cost recaptured with Python automation

~85% Efficiency Gain

Hours spent per week on manual Excel / reporting: 10 hrs / wk

1 hr (Minor task)   20 hrs (Part-time)   40 hrs (Full-time role)

Estimated team hourly rate / loaded cost: $30/hr

+ Custom rate

Projected Automation Impact   Annualized

HOURS SAVED / YEAR	ESTIMATED ANNUAL SAVINGS
~442 hrs	$13,260
(~55 full work days)	Direct labor recapture

DIRECT COMMUNICATION CHANNELS

ngwaradkar@gmail.com   +91-8668634502

Pune, Maharashtra, India   ● Available for consulting

Request Feasibility & Automation Quote

24h Response

Share your current manual files or workflow bottlenecks. Your ROI projection is automatically attached to help evaluate solution scope.

Your Name *   Work Email *

e.g. Anand Kulkarni   anand@company.com

Process / Workflow Subject *

Brief Process Description & Current Bottlenecks *

Describe your current manual steps (e.g. daily copy-paste from SAP, VLOOKUP formula lag, multi-file reconciliation) and desired automated deliverable format...

ROI Attached: 10 hrs/wk → ~442 hrs/yr ($13,260/yr)

Get Feasibility & Automation Quote

100% Confidential • Direct Developer Review   No Spam Guarantee

GitHub   ngwaradkar@gmail.com